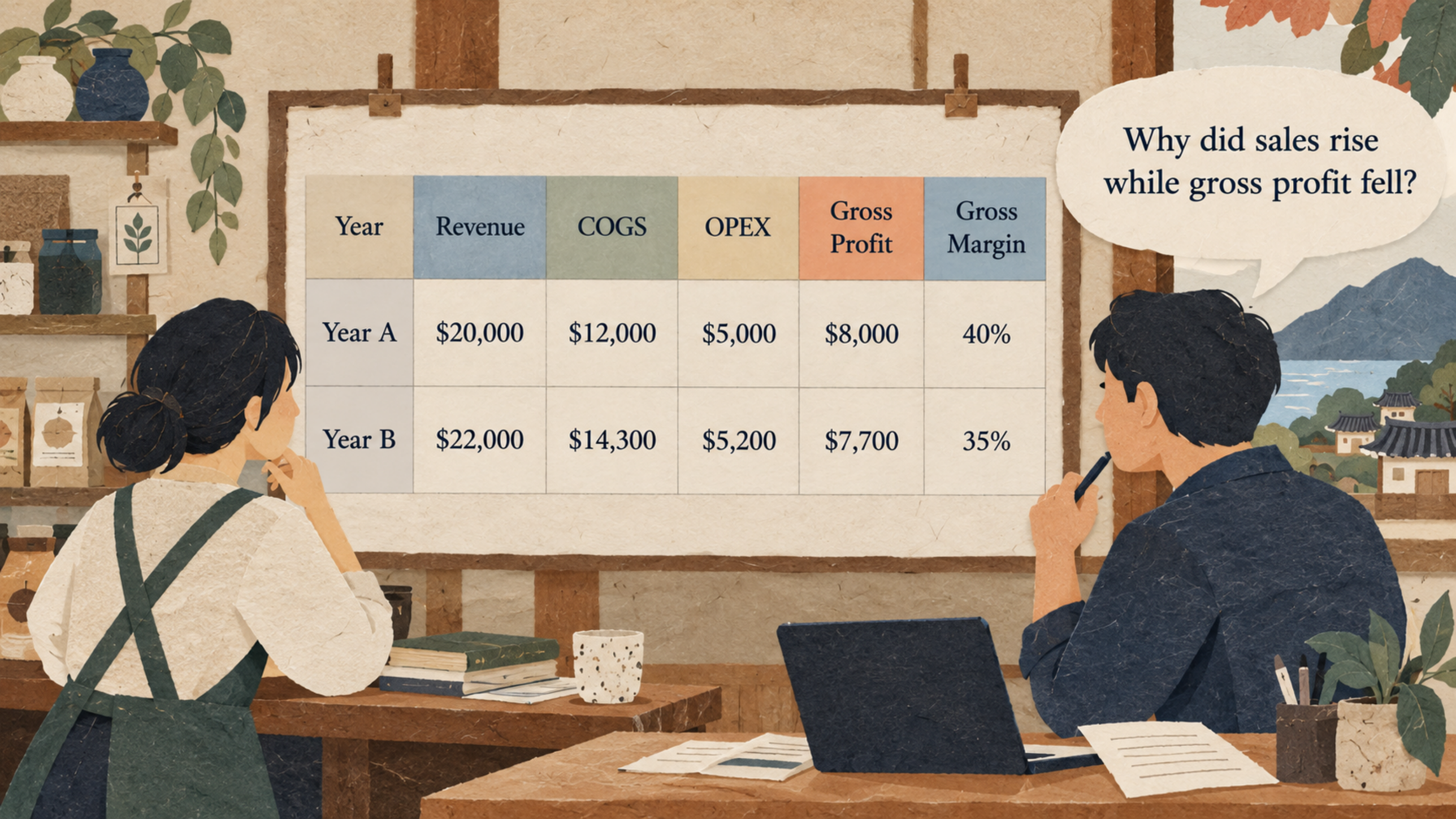


Year	Revenue	COGS	OPEX	Gross Profit	Gross Margin
Year A	$20,000	$12,000	$5,000	$8,000	40%
Year B	$22,000	$14,300	$5,200	$7,700	35%

What problem are we solving?

Learn how to use AI to read a simple income statement: give the AI the numbers and the formulas, check the results yourself, and ask for a short brief that separates facts from guesses.

THE SITUATION

The problem

Hana runs a small tea and snack shop. Her sales grew from $20,000 to $22,000, a 10% increase. But at the end of the year, she has less money left than before. Her root problem is that she does not know why, and she is about to hire a part-time worker. Her idea is to ask ChatGPT. She asked, "My sales went up 10%. Is my shop doing better? Should I hire?" ChatGPT said, "Yes. 10% growth is strong. Hiring will help you grow." But the shop's profit actually fell. The AI looked only at sales because the prompt did not include the costs.

Today's goal and a three-step method

Learn how to use AI to read a simple income statement: give the AI the numbers and the formulas, check the results yourself, and ask for a short brief that separates facts from guesses.

STEP 1

Give the AI the numbers and the formulas

Paste the data and define each measure, so the AI calculates the right thing.

STEP 2

Get a table, then check two cells

Ask for one table with all measures. Check at least two cells with a calculator.

STEP 3

Ask for a brief with facts and guesses separated

Ask for a short brief that states the facts, labels any explanation as "Not verified," and asks one question.

Hana's shop results (USD)

Fictional data created for this class. File: data/session_3_1_shop_results.csv

Year	Revenue	Cost of goods sold (COGS)	Operating expenses (OPEX)
Year A	20,000	12,000	5,000
Year B	22,000	14,300	5,200

Six money words you need first

Before you ask an AI about profit, you must know what each number means. Each example uses Year A.

KEY IDEA

Revenue

All the money customers paid. Year A: $20,000.

KEY IDEA

Cost of goods sold (COGS)

What the shop paid for the tea and snacks it sold. Year A: $12,000.

KEY IDEA

Gross profit

Revenue minus COGS. Year A: $20,000 - $12,000 = $8,000.

Six money words you need first (continued)

Before you ask an AI about profit, you must know what each number means. Each example uses Year A.

KEY IDEA

Gross margin

Gross profit as a percent of revenue. Year A: $8,000 / $20,000 = 40%. It shows how much of each dollar of sales is left after paying for the goods.

KEY IDEA

Operating expenses (OPEX)

The other costs of running the shop, such as rent and wages. Year A: $5,000.

KEY IDEA

Operating profit

Gross profit minus OPEX. Year A: $8,000 - $5,000 = $3,000. This is closer to the money the shop really keeps.

Why the AI answer was wrong

"Sales went up" and "the shop earns more" are different questions. Compare the two years.

Measure	Year A	Year B	Change
Revenue	$20,000	$22,000	+$2,000 (+10%)
Gross profit	$8,000	$7,700	-$300
Gross margin	40%	35%	-5 percentage points
Operating profit	$3,000	$2,500	-$500

KEY POINTS

What the table shows

- Sales went up by 10%, but operating profit went down by $500.
- COGS grew faster than revenue, so each dollar of sales left less money: 40 cents in Year A and 35 cents in Year B.
- A percentage point is the simple difference between two percents: 40% - 35% = 5 percentage points.

How the three-step method fixes the answer

The instructor runs Poor Prompt 1 and Improved Prompts 1 and 2 below in front of the class and compares the answers.

KEY IDEA

Step 1 fixes the missing-cost problem

With COGS, OPEX, and the formulas, the AI can see that profit fell.

KEY IDEA

Step 2 fixes the trust problem

AI tools can make arithmetic mistakes, especially with percents. Checking two cells takes one minute.

KEY IDEA

Step 3 fixes the guessing problem

The data shows that profit fell, not why it fell. Higher supplier prices or more discounts are possible explanations, so they must be labeled "Not verified."

Poor prompt 1

Same case. Compare what the prompt asks for.

POOR PROMPT 1 - DO NOT USE

AI Chat

You

My shop's sales went up from $20,000 to $22,000 this year.
Is my shop doing better? Should I hire a part-time worker?

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

"Yes. 10% revenue growth is a strong result. Hiring a part-time worker can help you serve more customers and keep growing."

WHY IT IS WEAK

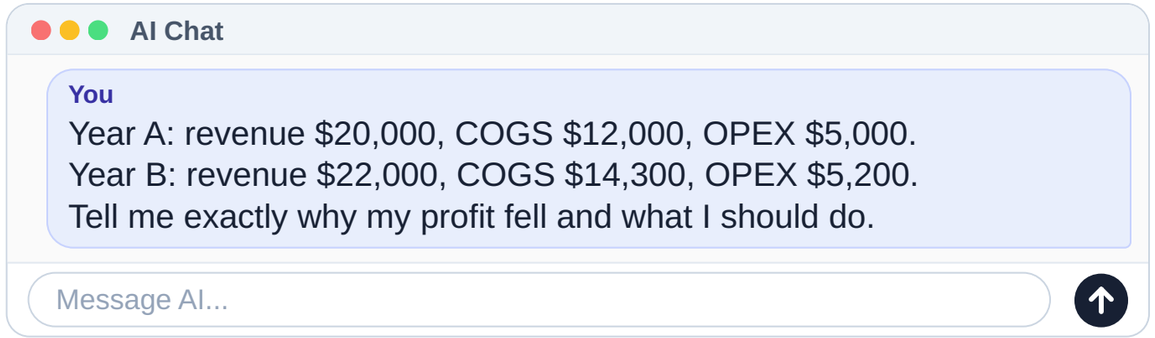
Reasons

- It gives only revenue, so the AI cannot see the costs or the profit.
- It asks a yes/no business question before any numbers are checked.
- The result is a confident "yes" to a decision that would make the shop's profit even smaller.

Poor prompt 2

Same case. Compare what the prompt asks for.

POOR PROMPT 2 - DO NOT USE



WHAT THE AI OFTEN ANSWERS

A typical result

"Your profit fell because your supplier raised prices. You should change suppliers and raise your prices by 10%."

WHY IT IS WEAK

Reasons

- It gives the numbers but no formulas, so the AI may calculate "profit" in a different way.
- "Tell me exactly why" asks for a cause that the data does not show, so the AI invents one.
- It asks for an action plan before the numbers are checked.

Improved prompt 1

Steps 1 and 2. Copy this prompt exactly.

IMPROVED PROMPT 1 - COPY AND SEND

AI Chat

You

I run a small tea and snack shop. Here are two years of results in USD.
Year A: revenue 20,000; cost of goods sold (COGS) 12,000; operating expenses (OPEX) 5,000.
Year B: revenue 22,000; COGS 14,300; OPEX 5,200.

Use these definitions:
- Gross profit = revenue - COGS
- Gross margin = gross profit / revenue, as a percent
- Operating profit = gross profit - OPEX

Make one table with a row for each year and these columns: revenue, COGS, gross profit, gross margin, OPEX, operating profit.
Add a last row that shows the change from Year A to Year B.
Show the calculation for Year B gross margin so I can check it.
Do not explain causes and do not recommend anything yet.

Message AI...

Why it works: improved prompt 1

Each part of the prompt matters.

WHY IT WORKS

Reasons

- It gives all the numbers and the formulas, which fixes the missing costs in Poor Prompt 1.
- The definitions make sure "profit" means the same thing to you and the AI.
- One table with a change row makes the drop in profit easy to see.
- It asks to see one calculation, so you can check the math.
- "Do not explain causes" keeps calculating and explaining as separate steps.

Improved prompt 2

Same case. Compare what the prompt asks for.

IMPROVED PROMPT 2 - COPY AND SEND

AI Chat

You

I checked your table. Year B gross profit is $7,700, gross margin is 35%,
and operating profit is $2,500, down $500 from Year A.

Write a brief for Hana, the shop owner, in under 100 words:
1. Two facts from the table, with numbers.
2. One possible explanation, labeled "Not verified."
3. One question Hana should check before deciding to hire.
Do not recommend hiring or not hiring yet.

Message AI...

WHY IT WORKS

Reasons

- It gives the AI the checked numbers, so the brief cannot change them.
- Two facts with numbers keep the brief grounded in the data.
- The "Not verified" label separates a guess from a fact.
- Asking for one question turns the guess into something Hana can check.

Exercise 1: Read the results

Guided practice. Follow the steps in order.

GUIDED PRACTICE

Steps

1. Open ChatGPT and start a **new chat**. Copy Poor Prompt 1 from the box above, paste it, and send it. Write down the AI's answer to "Should I hire?" You will need it for your report.
2. Start another **new chat**, so the first answer does not affect the second. Copy Improved Prompt 1, paste it, and send it.
3. Check two cells with a calculator: Year B gross profit is $22,000 - $14,300 = $7,700, and Year B gross margin is $7,700 / $22,000 = 35%. Compare the whole AI table with the answer key below.
4. In the same chat, copy Improved Prompt 2, paste it, and send it. Check that the explanation is labeled "Not verified" and that the brief does not recommend hiring.
5. For the Results section of your report, write the AI answer from Step 1 and one sentence that explains why it was wrong.

Answer key

Use this table to check the AI.

Year	Revenue	COGS	Gross profit	Gross margin	OPEX	Operating profit
Year A	$20,000	$12,000	$8,000	40%	$5,000	$3,000
Year B	$22,000	$14,300	$7,700	35%	$5,200	$2,500
Change	+$2,000	+$2,300	-$300	-5 points	+$200	-$500

Exercise 2: Check a new year and a hiring plan

Applied practice with a template.

HOW IT WORKS

Read first

- Hana now has Year C results, and she still wants to hire. Use this template. Replace each part in square brackets.

TEMPLATE - FILL IN, THEN SEND

AI Chat

You

Continue with the same definitions and the same table.
Here are the new numbers: [year and numbers].
Add a row for [year] and a change row compared with [earlier year].
Show the calculation for [one measure] so I can check it.
Then answer in under 80 words: [the question]. Give two facts with numbers,
label any explanation "Not verified," and end with one question to check.

Message AI...

Part A: Worked example (run it)

Copy and send this completed prompt in the same chat.

PART A - COPY AND SEND

AI Chat

You

Continue with the same definitions and the same table.
Here are the new numbers: Year C revenue 23,000; COGS 14,950; OPEX 5,300.
Add a row for Year C and a change row compared with Year B.
Show the calculation for Year C gross margin so I can check it.
Then answer in under 80 words: Did the shop improve from Year B to Year C?
Give two facts with numbers, label any explanation "Not verified," and end with one question to check.

Message AI...

CHECK THE ANSWER

Expected result

Year C gross profit is $23,000 - $14,950 = $8,050, and gross margin is $8,050 / $23,000 = 35%, the same as Year B. Operating profit is $8,050 - $5,300 = $2,750, up $250 from Year B. The shop improved a little, but the margin did not recover to 40%.

Part B: Your turn (write it)

Fill in the template yourself and send it.

YOUR TASK

Write your own prompt

A part-time worker would cost $3,000 a year, which adds to OPEX. Fill in the template yourself to ask whether Hana can hire in Year C. Your prompt must give the new OPEX and ask for the new operating profit.

CHECK THE ANSWER

Expected result

With the worker, Year C OPEX is $5,300 + $3,000 = $8,300. Operating profit is $8,050 - $8,300 = -$250, a loss. A good answer says that hiring now would turn the profit into a loss, unless sales or the margin grow.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W9_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- **Introduction:** Explain Hana's root problem (sales grew 10% but less money is left), her plan to hire, and why the first AI answer was wrong.
- **Method:** Copy Poor Prompt 1, Improved Prompts 1 and 2, the Part A prompt, and the Part B prompt you wrote.
- **Results:** Include the AI answer to Poor Prompt 1, the AI table with your check of two cells, the brief from Improved Prompt 2, and your checks for Year C and the hiring plan.
- **Conclusion:** Recommend whether Hana should hire now, in one or two sentences. Name one fact that supports it and one thing that is still not verified.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- You can explain in one sentence why "sales went up 10%" did not mean the shop was doing better, and your Part B prompt produces an operating profit of -$250 that you checked with a calculator.